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SUBNET PROFILE · Issue 006, Decentralized science

METANOVA: DeSci and DeAI Pave the Way for Medicine’s New Frontier

How Subnet 68, METANOVA, turns Bittensor into an "in silico" laboratory, screening molecular libraries for viable drug candidates in days instead of decades.

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Way for Medicine’s New Frontier

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The Human Genome Project (1990–2003) stands as one of the most ambitious and collaborative scientific endeavors in history, uniting laboratories worldwide to sequence and map the human genome, alongside those of several well-studied organisms. This monumental effort laid the foundation for modern biology, catalyzing rapid advancements in gene mapping, sequencing, and analysis.

The rise of bioinformatics, an interdisciplinary field, leveraging computer science, mathematics, and information technology to interpret biological data, has since transformed these processes. However, traditional drug discovery pipelines have struggled to improve the speed or efficacy of bringing new therapies to market. Despite progress in proteomics [the study of proteins], genomics [the study of genes], and pharmacology [the study of drugs], the journey from discovery to implementation often spans over a decade, costs billions of dollars, and succeeds in only about 10% of candidates reaching clinical trials.

Enter aMETANOVA_labs (Subnet 68, or SN68), launched on March 1, 2025, by Metanova Labs, a crypto-native biotech company powered by the Bittensor network. METANOVA represents a paradigm shift, harnessing decentralized artificial intelligence (AI) to accelerate drug discovery. Operating as an "in silico" [virtual, computer-based] laboratory, it tackles global health challenges by distributing computational tasks across a network of miners and validators, fundamentally altering the traditional model.

Revolutionizing Drug Discovery with Decentralized AI

Conventional drug discovery is a lengthy, siloed process with a success rate of approximately 1 in 10,000 candidates. METANOVA, by contrast, leverages machine learning to screen molecular libraries for small molecules with high affinity [ability to attach well] and synthesizability [ability to be created in a lab] in mere days, at a fraction of the cost. This is achieved through a decentralized science (deSci) framework and

incentive mechanisms on Bittensor, rewarding participants for innovative contributions.

Initially, METANOVA's Version 1 (V1) focused on predicting "binding affinity" to specific protein targets using PSICHIC, a graph-based model that assesses binding potential from sequence data alone. This approach enables rapid screening of hundreds of thousands of compounds per hour, a significant leap from traditional methods.

Unlike the competitive silos of conventional research, METANOVA's Bittensor framework fosters collaboration. Miners stress-test open-source models, identifying weaknesses that validators use to refine predictions, ensuring continuous improvement, a process aligned with the subnet's merit-based design.

Scaling Complexity and Enhancing Precision

As METANOVA scales, it addresses increasingly complex challenges, incorporating multiple targets, physicochemical properties, and additional metrics such as pharmacokinetics [how a drug behaves in the body] and toxicity. A molecule may bind strongly to a disease-related protein but fail as a therapy if it lacks these properties. METANOVA aims to predict such outcomes, expanding its validation process beyond binding affinity.

The April 2025 "Shannon upgrade" marked a significant evolution, introducing a sophisticated scoring mechanism, dynamic chemical mapping, and a merit-based adversarial framework. This upgrade enhanced molecular diversity through multi-molecule challenges, a target–antitarget evaluation system, and a "Shannon diversity bonus" rewarding chemically diverse submissions.

Version 2 (V2) further refined the platform, adopting a model-agnostic design to integrate future predictive tools beyond PSICHIC, ensuring adaptability as the field evolves.

Recent Innovations and Industry Alignment

In August 2025, Metanova Labs announced Phase I and Phase II updates, informed by feedback from pharmaceutical partners, academia, community and Bittensor leaders including @const_reborn. Phase I integrates Boltz-2, a structure-based model co-developed by MIT and Recursion Pharma, which predicts protein–molecule interactions with physics-level accuracy at a fraction of traditional computational cost (1000x faster than free-energy perturbation). Paired with PSICHIC, this dual approach offers rapid screening and detailed validation, accelerating the path from hit identification to lead optimization.

Phase II shifts the focus from raw compute power to efficient search strategies, capping compute and time budgets to incentivize algorithmic innovation. Miners now submit code for standardized evaluation, democratizing participation and reducing reliance on hardware. This phase also enhances intellectual

property (IP) safety, allowing private data under confidential disclosure agreements (CDAs), addressing pharma industry sensitivities.

Recent updates, including the addition of Suzuki-Miyaura coupling to the combinatorial dataset (unlocking 17 billion+ possibilities as of September 2025), underscore METANOVA's commitment to expanding its chemical

libraries.

Future Prospects and Ecosystem Impact

METANOVA envisions evolving into a network of specialized AI modules, extending beyond activity predictions to include medicinal chemistry filters for safety and efficacy. The potential introduction of sub-subnets could further diversify its scope, integrating insights to guide breakthrough strategies.

The broader deSci movement, driven by demands for transparency, is gaining traction, with the AI in drug discovery market projected to grow at a 29.7% compound annual growth rate from 2024 to 2030 (Grand View Research, 2023). METANOVA, undervalued relative to its potential, disrupts a trillion- dollar industry traditionally opaque to the outside world. While its full impact may only be recognized when a pharmaceutical giant or renowned lab partners with Metanova Labs, its trajectory suggests a future where decentralized AI could render conventional pipelines obsolete.

Conclusion

METANOVA stands at the forefront of a new frontier in medicine, blending decentralized science and AI to address global health challenges. Its iterative approach, grounded in scientific rigor and community collaboration, positions it as a blueprint for the future of drug discovery—one that promises faster, more cost-effective, and transparent solutions for humanity.

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Simplifying Bittensor subnet ecosystem through accessible, insightful articles. #TAO